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IMPACT MILL AND A RESIDUE PROCESSING SYSTEM INCORPORATING SAME

Abstract

Described herein is a combine harvester having at least one impact mill for devitalising weed seeds in a crop while the crop is being harvested by the combine harvester and a power take off shaft. The impact mill of the combine harvester includes an inlet for material to enter the mill, an impact mechanism arranged to rotate about a rotation axis, an outlet for discharge of impacted material, a drive system coupled to the power take off shaft, one or more torque sensors, and a data processor arranged to process information provided by the one or more torque sensors wherein the data processor is programmed to calculate throughput of material processed by the impact mill. Also described herein is an impact mill system capable of mounting on a combine harvester for devitalising weed seeds in a harvested crop.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/040,090, filed Sep. 22, 2020, which is a national stage entry of International Application No. PCT/AU2019/050260, filed Mar. 22, 2019, which claims priority to AU201800350A, filed Mar. 22, 2018, and AU2018100351A, filed Mar. 22, 2018, each of which are incorporated by reference herein in their entirety.

TECHNICAL FIELD
[0002] An impact mill of a type suitable for the devitalisation of weed seeds and fragmentation of organic matter is disclosed. The impact mill in a preferred form is a multistage hammer mill. Also disclosed is a residue processing system which incorporates one or more impact mills or alternate residue processing devices.
BACKGROUND ART
[0003] Weeds and weed control are, and always have been, one of the biggest constraints and costs to grain production. Weeds are a perpetual problem that limits the food production capacity of agricultural area around the globe. Weeds compete with the cultivated crops for water, sunlight and nutrients. In the past 50 years there has been a shift from tillage being the most important method to control weeds to herbicides being the most important tool to control weeds. Herbicides in general provide much better control of weeds than tillage methods and do not have the major issues of soil erosion, moisture loss and breakdown of soil structure. The wide spread use and reliance of herbicides has resulted weeds evolving resistance to herbicides. The herbicide resistance is now widespread and presents one of the biggest threats to global food security. Strategies to provide non-chemical weed control to compliment herbicides are now paramount to reduce the selection pressure for herbicide resistance. One particular method of significant renewed interest is destroying weed seeds at harvest time to interrupt the weed cycle.
[0004] Many in crop weeds share a similar life cycle to harvested crops. Once a crop matures and is harvested, there is a broad range of weeds that have viable seeds remaining on the plant above the cutting height of the harvester. These weeds enter the harvester and their seeds either end up in a grain tank, out with straw residues, or out with chaff residues. There are a range of factors that determine where a weed seed will end up at harvest time including moisture content, maturity, and harvester setup. A major factor that determines where a seed ends up is the aerodynamic properties of the seeds or its terminal velocity. Often a weed seed is much lighter than the grain being harvested. Crop cleaning system used during harvesting employ a winnowing action to remove light chaff material from the heavier grain using airflow and mechanical sieving. The light weed seeds are caught in the wind and can exit the back of the harvester sieve. The residues and contained weed seeds are then spread on the ground to be a problem for next year. The residues also contain a proportion of grain being harvested that could not be separated by the harvester. This

grain loss has the potential to become a volunteer weed after harvest. There is an opportunity to intercept and destroy weed seeds in the residues before allowing them to become a problem for next year's crop.
[0005] One method to destroy these weed seeds is to use a milling technology. Milling technology has been used for particle size reduction of a range of feedstock for over a century. Milling technology can be separated into crushing and impact technology.
[0006] The most common crushing size reduction technology is the roller mill. Roller mills have been investigated for the purpose of destroying weed seeds at harvest time. Roy and Bailey (1969) U.S. Pat. No. 3,448,933 describe a roller shear mill for destroying weed seeds out of clean grain screenings. Reyenga (1991) U.S. Pat. No. 5,059,154 describes using a separating device and roller mill to crush foreign matter such as weed seeds. A limitation of the roller mill is the ability to handle the bulk of residue material that contains the weed seeds and thus rely on a separation means to reduce the residue material.
[0007] Impact mills use high impact speeds generated by rotating elements to pulverise material. Impact mills have also been of interest for the destruction of weed seeds at harvest.
[0008] A widely used type of impact mill is a hammer mill, which uses a rotor with impact elements to pulverise material and a screen to classify the output size distribution. Hammer mills are highly versatile and are able to accept a wide range of feed materials. Plant material such as crop residues is fibrous and difficult to process. The use of hammer mills to devitalise weed seeds in crop residues has been well documented. The use of hammer mills on board a harvester to devitalise weed seeds has been subject of multiple patents (e.g. Wallis (1995) AU1996071759 Bernard (1998) FR2776468B1)).
[0009] An advantage of hammer mills is that in addition to impact, they induce crushing, shear and attrition forces that make them particularly useful for size reduction of fibrous materials. Another advantage of hammer mills is that they often have flexible impact elements that are replaceable and can handle some foreign objects without damage.
[0010] A further advantage of the hammer mill is that the screen size controls particle fineness and can then control the proportion of weed devitalisation. Control of output size distribution is particularly valuable in the processing of crop residues where material type and moisture conditions change significantly. Change in material conditions result in still similar output size distribution and weed seed devitalisation remains less dependent on material conditions than would be without the use of screens.
[0011] A disadvantage of current hammer mills is that the screen which controls particle size distribution determines throughput capacity. In general, to devitalise weed seeds a small screen size is required and hence throughput capacity is limited. A hammer mill with concentric screens of varying sizes has been described by Emmanouilidis (1951) U.S. Pat. No. 2,557,865. The Emmanouilidis mill has a central impact zone and additional screens are used to separate output material into different size fractions. The inner primary zone in the Emmanouilidis mill still dictates capacity and overall size reduction.
[0012] A different type of impact mill is a cage mill. A cage mill applies predominantly impact forces and level of size reduction is set through rotational speed and the number of concentric rows of bars. There is no classification of particle size with a cage mill. The impact forces in a cage mill make them suitable for friable or brittle materials and are not widely used for processing fibrous materials. However, one example is described in AU 2001/038781 (Zani) which is proposed for destruction of weed seeds. The Zani cage mill has concentric rows of impact elements supported by a ring. The mill is driven at high impact speed to destroy weed seeds. The arrangement can be neatly integrated into the harvester. The arrangement however has limited capacity and cannot process the entire chaff residue fraction exiting the harvesters sieve. Therefore, the Zani system relied on sieving to concentrate the collect weed seeds for processing.
[0013] An increased capacity cage mill is described in WO 2009/100500 (Harrington) to handle the

whole chaff material fraction to destroy weed seeds. The Harrington used a large counter rotating cage mill that has fan blades similar to Tjumanok et al 1989 (U.S. Pat. No. 4,813,619) to increase airflow and capacity. This cage mill is large, heavy, requires a complex counter rotating drive and requires considerable power to operate. The system has its own power package and is towed behind the grain harvester. The size, weight and drive, limits options to integrate the cage mill into the harvester. The mill incorporates cylindrical bars that limit impact speeds because of glancing blows. The impact speed therefore has a large distribution. To get sufficient impact energy into weed seeds requires counter rotation of the cage structures.

[0014] The current state of the art for seed destroying mill technology is described in PCT/AU2014/218502 (Berry Saunders). Berry Saunders uses a rotor stator cage mill that is much simpler to integrate into a grain harvester than the counter rotation systems. The Berry Saunders mill provides an advance on the Zani cage mill by improving the throughput capacity and seed kill performance of the mill system. It achieves this by using a central distribution element (also described in Isaak (2003) DE 10203502) and angular static bars that are slanted against the rotation of the rotor. A purportedly novel aspect of Berry Saunders mill is the spacing between the angled impact bars determines if a seed will pass through to the next row of impact bars or stay within the current row of impact bars. The size of the seed does not determine if it passes through the row of impact bars or remains.

[0015] The relatively simple workings of cage mills which apply predominantly impact and do not use size classification has enabled computer modelling techniques to be used to predict mill performance. The Berry Saunders mill has been optimised using computer modelling techniques to apply the ideal requirements to devitalise weed seeds using impact alone. However, there has been little concern for the airflow component of the power consumption. The rotor bars are narrow with sharp edges resulting in high drag coefficient and turbulence generation. The stator bars are orientated to result in torque converter or water brake dynamometer like turbulence generation and wasted heat generation.

[0016] One disadvantage of this approach is that the stator impact bars take up a lot of space radially. This in turns means that adjacent rows of rotating impact bars are spaced a long way apart. For a weed seed devitalisation mill, or a particle destruction mill for that matter impact speed is crucial. When impact bars are spaced widely apart the impact speed difference between each subsequent row is significant.

[0017] The above references to the background art do not constitute an admission that the art forms a part of the common general knowledge of a person of ordinary skill in the art. The above references are also not intended to limit the application of the method and system as disclosed herein.

SUMMARY OF THE DISCLOSURE

[0018] In a first aspect there is disclosed a multistage hammer mill comprising: [0019] a plurality of milling stages arranged concentrically about each other; the plurality of milling stages arranged so that substantially all material in a first inner most of the milling stages passes through at least one subsequent adjacent milling stage, the plurality of milling stages including a first milling stage and a second milling stage, a central feed opening enabling material flow into a primary impact zone of the first milling stage; [0020] the first milling stage comprising an impact mechanism and a first screen arrangement, the impact mechanism located in the primary impact zone and arranged to impact material entering the primary impact zone and accelerate the impacted material in a radial outward direction, the impact mechanism being capable of rotating about a rotation axis, the first screen arrangement disposed circumferentially about and radially spaced from the impact mechanism the first screen arrangement being provided with a plurality of apertures through which impacted material of a first size range can pass; [0021] the second milling stage comprising a second arrangement disposed circumferentially about and radially spaced from the first screen arrangement, the second screen arrangement being provided with a plurality of apertures through

impacted material of a second size range can pass, the second size range being the same as or different to the first size range, and [0022] one or more impact elements disposed between the first screen arrangement and the second screen arrangement, wherein material entering the second milling stage from the first milling stage is impacted and accelerated by the impact elements and pulverised against the screen arrangement.
[0023] In a second aspect there is disclosed a residue processing system for an agricultural machine having a power source with a power take off rotating about a first axis, the residue processing system comprising: at least one residue processing device each having a respective first drive shaft rotatable about a respective axis perpendicular to the first axis; a transmission system coupled between the PTO and each first drive shaft to change a direction of drive from the PTO to each first drive shaft and a belt drive arrangement coupled between the transmission system and each first drive shaft to transfer torque from the PTO to each first drive shaft.
[0024] In a third aspect there is disclosed a combine harvester comprising: a power take off (PTO) rotating about a power axis perpendicular to a direction of travel of the combine harvester; least one multistage hammermills according the first aspect, each hammer mill having at least a first drive shaft for imparting rotation to the impact mechanism of the respective hammer mills about respective axes perpendicular to the power axis; [0025] a transmission system arranged to change a direction of drive from the PTO to each first drive shaft; and a belt drive arrangement coupled between the transmission system and each first drive shaft to transfer torque from the PTO to each first drive shaft.
[0026] In a fourth aspect there is disclosed an impact mill comprising: [0027] an inlet for material to enter the mill, one or more rotating elements arranged to rotate about a rotation axis, the rotating elements being operable to pulverise the material after entering through the inlet, and an outlet for discharge of pulverised material; and one or more blockage sensors arranged detect blockage in or reduced mass flow rate of material through the mill.
[0028] In a fifth aspect there is disclosed an impact mill comprising: [0029] an inlet for material to enter the mill, one or more rotating elements arranged to rotate about a rotation axis, the rotating elements being operable to pulverise the material after entering through the inlet, and an outlet for discharge of pulverised material; and [0030] one or more vibration sensors arranged to sense vibration arising from rotation of the impact mechanism about the rotation axis.
[0031] In a sixth aspect there is disclosed an impact mill comprising: [0032] an inlet for material to enter the mill, an impact mechanism arranged to rotate about a rotation axis, the impact mechanism being operable to pulverise the material after entering through the inlet, and an outlet for discharge of pulverised material; and [0033] (a) one or more blockage sensors arranged detect blockage in or reduced mass flow rate of material through the mill; or [0034] (b) one or more vibration sensors arranged to sense vibration arising from rotation of the impact mechanism about the rotation axis; or [0035] (c) one or more blockage sensors arranged detect blockage in or reduced mass flow rate of material through the mill and one or more vibration sensors arranged to sense vibration arising from rotation of the impact mechanism about the rotation axis.
[0036] In one embodiment the one or more blockage sensors are provided to detect: [0037] (a) a blockage at or reduced mass flow into the inlet of the mill; and/or [0038] (b) a blockage at or reduced mass flow rate from the outlet of the mill.
[0039] In one embodiment the impact mill comprises at least one temperature sensor arranged to provide a temperature measurement of a bearing on which the impact mechanism rotate.
[0040] In one embodiment the impact mill comprises, or is arranged to be, operatively associated with a data processor in communication with the at least one temperature sensor wherein the data processor is arranged to: [0041] (i) issue an alarm or warning to an operator of the mill of the temperature measurement being above a threshold; or [0042] (ii) autonomously reduce the speed or stop the rotation of the impact mechanism; or [0043] (iii) activate a mill cooling system for cooling the bearing; or [0044] (iv) any combination of two or more of (i), (ii) and (iii); [0045] when the

temperature measurements exceed the threshold.
[0046] In one embodiment the data processor is in communication with the one or more vibration sensors, the data processor being arranged to process signals from the vibration sensors and: [0047] (a) issue a warning or alarm to an operator of the mill; or [0048] (b) facilitate an autonomous reduction in rotational speed of the mill; or [0049] (c) facilitate a shut down the mill to a rotational speed of 0 rpm; or [0050] (d) issue a warning or alarm to an operator of the mill and facilitate an autonomous reduction in rotational speed of the mill; or [0051] (e) issue a warning or alarm to an operator of the mill and facilitate a shut down the mill to a rotational speed of 0 rpm; [0052] if the processed signals are indicative of mill vibration being in excess of a threshold.
[0053] In a seventh aspect there is disclosed a combine harvester comprising: [0054] at least one impact mill according to any one of claims **1-3** and a data processor arranged to process information provided by the one or more blockage sensors wherein the data processor is programmed to: [0055] (i) direct one or more high-pressure jets of air to the inlet and/or the outlet when an anomalous change in material flow, or a blockage is detected; or [0056] (ii) reduce ground speed of the combine harvester when an anomalous change in material flow through the mill, or a blockage of the mill detected; or [0057] (iii) both (i) and (ii)
[0058] In one embodiment the combine harvester comprises a chopper arranged to receive a discharge flow from the at least one multistage hammer mill; and one or more additional blockage sensors arranged detect a blockage in or reduced mass flow rate of material through the chopper.
[0059] In one embodiment of the combine harvester the data processor is arranged to: [0060] (1) issue an alarm or warning to an operator of the mill of the temperature measurement being above a threshold; or [0061] (2) autonomously reduce the speed or stop the rotation of the impact mechanism; or [0062] (3) activate a mill cooling system for cooling the bearing; or [0063] (4) any combination of two or more of (1), (2) and (3); [0064] when the temperature measurements exceed the threshold.
[0065] In one embodiment of the combine harvester the data processor is arranged to: [0066] (a) issue a warning or alarm to an operator of the mill; or [0067] (b) facilitate an autonomous reduction in rotational speed of the mill; or [0068] (c) facilitate a shut down the mill to a rotational speed of 0 rpm; or [0069] (d) issue a warning or alarm to an operator of the mill and facilitate an autonomous reduction in rotational speed of the mill; or [0070] (e) issue a warning or alarm to an operator of the mill and facilitate a shut down the mill to a rotational speed of 0 rpm; [0071] if the processed signals are indicative of mill vibration being in excess of a threshold.
[0072] In an eighth aspect there is disclosed an impact mill comprising: [0073] an inlet for material to enter the mill, an impact mechanism arranged to rotate about a rotation axis, the impact mechanism being operable to pulverise the material after entering through the inlet, and an outlet for discharge of pulverised material; and [0074] one or more torque sensors arranged to provide a signal indicative of torque or load applied to a drive shaft of the impact mechanism.
[0075] In one embodiment each torque sensor may comprise first and second encoders spaced apart along with the drive shaft. In such embodiment the data processor may be programmed or otherwise arranged to calculate or provide indication of torque on the basis of a difference between outputs of the first and second encoders. In one embodiment the first and second encoders are located at opposite ends of the drive shaft.
[0076] In each of the above described embodiments the data processor associated with the mill and/or the combine harvester may also include or be coupled with a communications system enabling data, signals or information from any one or more of the sensors (i.e. the vibration sensors, blockage sensors, temperature sensors, proximity sensors, and torque sensors) to be communicated via a communications network including but not limited to the Internet or the Internet of things, to a remote location. The data, signals or information from the sensors may be provided directly from the sensors, or, subsequent to processing by the data processor, or both.
[0077] Communicating the data, signals or information enables remote monitoring of the

performance of the mill and/or the combine harvester. The remote monitoring can for example enable manual or automated communication to a combine operator a service department of performance characteristics of the mill and/or the combine harvester. The performance characteristics may include information regarding wear of various components, the need for maintenance or the provision in real time of alerts or alarms to the combine operator of potentially dangerous performance characteristics such as bearing temperature.

[0078] The data, signals or information may also be used to calculate the amount of material processed by the combine harvester. This may be beneficial in terms of different business or revenue models for commercialisation of the mill all combine harvester in enabling for example lease payments/charges being made on the basis of the calculated amount of material processed by the combine harvester.

[0079] In one embodiment of the fourth to seventh aspects impact mill is a multistage hammer mill comprising: [0080] a plurality of milling stages arranged concentrically about each other; the plurality of milling stages arranged so that substantially all material in a first inner most of the milling stages passes through at least one subsequent adjacent milling stage, the plurality of milling stages including a first milling stage and a second milling stage; [0081] the inlet is a central feed opening enabling material flow into a primary impact zone of the first milling stage; [0082] the first milling stage has an impact mechanism and a first screen arrangement, the impact mechanism located in the primary impact zone and arranged to impact material entering the primary impact zone and accelerate the impacted material in a radial outward direction, the impact mechanism being capable of rotating about the rotation axis, the first screen arrangement disposed circumferentially about and radially spaced from the impact mechanism the first screen arrangement being provided with a plurality of apertures through which impacted material of a first size range can pass; [0083] the second milling stage has a second arrangement disposed circumferentially about and radially spaced from the first screen arrangement, the second screen arrangement being provided with a plurality of apertures through impacted material of a second size range can pass to flow toward the outlet, the second size range being the same as or different to the first size range; and [0084] one or more impact elements are disposed between the first screen arrangement and the second screen arrangement, wherein material entering the second milling stage from the first milling stage is impacted and accelerated by the impact elements and pulverised against the screen arrangement.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0085] Notwithstanding any other forms which may fall within the scope of the hammer mill as set forth in the Summary, specific embodiments will now be described, by way of example only, with reference to the covering drawings in which:

[0086] FIG. **1** is an isometric view of an embodiment of the disclosed impact mill as a multi stage hammer mill;

[0087] FIG. **2** is a section view of the multi stage hammer mill shown in FIG. **1** taken in a radial plane;

[0088] FIG. **3** is an isometric view of an arrangement of screens utilised in the embodiment of the multistage hammer mill shown in FIGS. **1** and **2**;

[0089] FIG. **4** is an isometric view of a screen structure which comprises the arrangement of screens shown in FIG. **3** coupled together with an inlet plate; and

[0090] FIG. **5***a* is an isometric view of a central impact mechanism and an arrangement of impact elements incorporated in an embodiment of the disclosed multistage hammer mill;

[0091] FIG. **5***b* is an enlarged section view of an impact elements incorporated in the disclosed

hammer mill;
[0092] FIG. **6** is an exploded view of the disclosed multistage hammer mill;
[0093] FIG. **7***a* is a schematic representation of a residue processing system which includes two of residue processing devices that have two counter rotating components in a side-by-side juxtaposition;
[0094] FIG. **7***b* is a schematic representation of a drive belt arrangement used in the residue processing system shown in FIG. **7***a;*
[0095] FIG. **8***a* is a schematic representation of a rear portion of a combine harvester incorporating an embodiment of the disclosed residue processing system which directs processed material into a straw chopping device of the harvester;
[0096] FIG. **8***b* is a plan view of two hammer mills and the chopper utilised in the combine harvester shown in FIG. **8***a;*
[0097] FIG. **9** is a schematic representation of a rear portion of a combine harvester incorporating an alternate embodiment of the disclosed residue processing system which directs processed material directly onto a tailboard for spreading chaff and straw material;
[0098] FIG. **10***a* is a schematic representation of an embodiment of the residue processing system incorporating a drive system for a residue processing device that has two devices with one rotating component each;
[0099] FIG. **10***b* is a plan view of a drive belt arrangement used in the residue processing system shown in FIG. **10***a;*
[0100] FIG. **11***a* is a schematic representation of an embodiment of the residue processing system incorporating a modified drive system in comparison to that shown in FIG. **10***a;*
[0101] FIG. **11***b* is a plan view of a used in the residue processing system shown in FIG. **11***a;*
[0102] FIG. **12** is a schematic representation of a further drive belt arrangement having a pulley incorporating a fan which may be used in embodiments of the residue processing system;
[0103] FIG. **13***a* is a schematic representation of a residue processing system having two of the disclosed hammermills juxtapose side-by-side and rotating in the same direction and with their respective covers off;
[0104] FIG. **13***b* is a representation of the hammermills shown in FIG. **13***a* but with their respective covers on;
[0105] FIG. **14** is a schematic representation of a further embodiment of the multistage hammer mill;
[0106] FIG. **15***a* is a schematic representation from the front of a segment of a screen arrangement provided with ribs on its radial inner surface which may be incorporated in embodiments of the disclosed multistage hammer mill;
[0107] FIG. **15***b* is a plan view of the segment shown in FIG. **15***a;*
[0108] FIG. **16** is a schematic representation of an alternate construction of screen arrangement incorporated in the disclosed multistage hammer mill;
[0109] FIG. **17** is a schematic representation of axial and radial scrapers the may be provided on the rings used for supporting the upper ends of the impact elements together in an alternate embodiment of the disclosed multistage hammer mill;
[0110] FIG. **18** is a schematic representation of a pulverising block which may be incorporated in alternate embodiments of the disclosed multistage hammer mill; and
[0111] FIG. **19** is a schematic representation of flow guide plates that may be incorporated in alternate embodiments of the disclosed multistage hammer mill.

DETAILED DESCRIPTION OF SPECIFIC EMBODIMENTS

[0112] With reference to the drawings there is shown an embodiment of the disclosed impact mill **10** (FIGS. **1**,**2** and **6**). The Impact mill **10** has an inlet **12** for material to enter the mill and an impact mechanism **16** arranged to rotate about a rotation axis **18**. The impact mechanism **16** is operable to pulverise the material after entering through the inlet **12** and discharges the pulverised material through an outlet **154** (FIG. **13***a*). As explained in greater detail later in this specification the impact mill **10** may include: [0113] (a) one or more blockage sensors Bj arranged detect blockage in or reduced mass flow rate of material through the mill; or [0114] (b) one or more vibration sensors V**1**, V**2** (FIG. **7***a*) arranged to sense vibration arising from rotation of the impact mechanism **16** about the rotation axis **18**; or [0115] (c) both one or more blockage sensors Bj and one or more vibration sensors V**1**, V**2**.

[0116] As seen in FIGS. **1**, **2** and **6** an embodiment of the disclosed an impact mill may be in the form of a multistage hammer mill **10** (hereinafter referred to in general "hammer mill **10**"). The central feed opening **12** allows material to flow into a primary impact or destruction zone **14**. The impact mechanism **16** is located in the primary impact zone **14** and is capable of rotating about the rotation axis **18**. The impact mechanism **16** is arranged to impact the material entering the primary impact zone **14** and accelerate the impacted material in a radial outward direction. The hammer mill **10** also has a first screen arrangement **20***a* and at least a second screen arrangement **20***b*. The first screen arrangement **20***a* is disposed circumferentially about the impact mechanism **16** and forms a boundary of the primary impact zone **14**. The first screen arrangement **20***a* has a plurality of apertures **22***a* through which impacted material of a first size range can pass.

[0117] The second screen arrangement **20***b* is disposed circumferentially about and radially spaced from the first screen arrangement **20***a*. The second screen arrangement **20***b* has a plurality of apertures **22***b* through which impacted material of a second size range can pass. The second size range can be the same as or different to the first size range. However in the present illustrated embodiment the second size range is different to the first size range. In particular a lower size limit of the second range is smaller than a lower size limit for the first range. The provision of the first and second screen arrangements **20***a* and **20***b* characterised the hammer mill **10** as being a two-stage hammer mill.

[0118] In this embodiment the hammer mill **10** is also provided with an optional third screen arrangement **20***c*. The third screen arrangement is disposed circumferentially about and radially spaced from the second screen arrangement **20***b*. The third screen arrangement **20***c* has a plurality of apertures **22***c* through which impacted material of a third size range can pass. The third size range can be arranged to have a lower limit that is the same or smaller than the lower limit of the second size range, although in this particular embodiment the lower limit is smaller for the third size range than the second size range.

[0119] The hammer mill **10** when provided with the third screen arrangement **22***c* constitutes a three stage hammer mill.

[0120] In the following discussion of the hammer mill **10** the first, second and third are screen arrangements are referred to in general as "screen arrangements **20**" and the apertures **22***a*, **22***b* and **22***c* are referred to in general as "apertures **22**". The apertures **22** are of a generally rectangular with rounded corners. As described above, the apertures **22** are of smaller size for the screen arrangements **20** with increased radius.

[0121] With reference to FIG. **2** it can be seen that the first screen arrangement **20***a* is formed with at least one (and in this particular embodiment three) openings or gaps **24***a*. The openings/gaps **24***a* are dimensioned to enable the passage of impacted material that is too large to otherwise pass through the apertures **22***a* in the screen **20***a*. This assists in minimising the build-up of oversized material within the primary impact zone **14** that may otherwise reduce the throughput of material through the hammer mill **10**. For example this may include pieces of straw or other plant matter which is entrained in the material fed into the hammer mill **10** through the feed opening **12**.

[0122] Likewise the second and third screen arrangements **20***b* and **20***c* may be provided with one or more (and this embodiment three) openings or gap **24***b* and **24***c* respectively to enable the passage of impacted material that is otherwise too large to pass through their respective apertures **22**. The gaps **24** also enable the passage of hard materials such as stones to minimise the risk of damage to the respective screen arrangements **20**.

[0123] The openings/gaps **24** of respective successive screen arrangements at least partially overlap in the circumferential direction. For example there is a circumferential overlap between the gaps **24***a* and **24***b*. Similarly there is a circumferential overlap between the gaps **24***b* and **24***c*.

[0124] When a screen arrangement **20** is formed with a plurality of openings/gaps **24** the openings/gaps **24** are evenly spaced circumferentially about the respective screen arrangement **20**.

[0125] In this embodiment the arc length of the respective gaps **24** increases with increased radius from the rotation axis **18**.

[0126] Each of the screen arrangements **20**, at least when provided with two or more openings/gaps **24**, may be formed from an identical number of screen segments **26**. The openings **24** are formed by appropriately circumferentially spacing apart the respective segments **26**. The number, spacing and relative position of the gaps **24** in mutually adjacent screen arrangements **20**, can be varied by changing the number and arc length of the respective segments **26** which make up each screen arrangement **20**. The relative position of the gaps **24** can also be varied by rotating the screen arrangements **20** relative to each other. Varying the position of the gaps **24** between adjacent screen arrangements **20** can effectively vary the maximum rotation of material about the respective screen arrangement prior to exiting to the next screen arrangement/stage.

[0127] A plurality of axially extending supporting ribs **28** is provided immediately behind each of the screen arrangements **20** in the radial direction. The ribs **28** are evenly spaced circumferentially about the respective screen arrangements **20**. The ribs **28** on a trailing side of each opening **24** with reference to the direction of rotation of the impact mechanism **16** may act as impact ribs **28***i* for material passing from one milling stage to the next. The impact ribs **28***i* also assist in slowing down hard materials flowing through the openings **24**.

[0128] Optionally for the third screen arrangement **20***c* at least one rib **28***g* is placed in each of the gaps **24***c*. The ribs **28***g* have the same shape and configuration as ribs **28** but acts as an impact bar for particles travelling through the opening **24***c*. The spacing of the rib **28***g* can increase with each outward screen arrangement and still provide effective impact for fragmenting material passing through the gaps **24***c* due to the increase in the tangential component of velocity relative to the radial component with increased radial distance from the rotation axis **18**. Evenly spacing the ribs **28***g* in the gaps **24***c* minimises the chance of material missing the ribs **28***g*. In addition to improving efficiency of fragmentation of the material, when the screen arrangements **20** are stationary, the ribs **28***g* assist in decelerating hard materials that may be entrained in the flow. This further reduces the likelihood of damage to the mill **10**. Also, in this regard the ribs **28***g* may be sacrificial to the extent that they are damaged in preference to the screen arrangement **20**.

[0129] With particular reference to FIG. **3** the axially opposite ends of the screen segments **26** of screen arrangement **20***a* are attached to the upper and lower rings **30***a*. The axially opposite ends of the screen segments **26** for screen arrangement **20***b* are attached to the upper and lower rings **30***b*. The axially opposite ends of the screen segments **26** for screen arrangement **20***c* are attached to the upper and lower rings **30***c*.

[0130] The screen arrangements **20** are fixed relative to each other by coupling to a common upper annular plate **32** shown in FIGS. **1** and **4**. This forms a screen structure **33**. The annular plate **32** is formed with a central opening which constitutes the feed opening **12**. The radius of the feed opening **12** is smaller than the radius of the first (i.e. inner most) screen arrangement **20***a*. This dimensional relationship facilitates acceleration of air and material in the radial outward direction as it enters the primary impact zone **14**.

[0131] Referring particular to FIGS. **2** and **5***a* the impact mechanism **16** is provided with a plurality (in this instance six) radially extending flails or hammers **34**. Each hammer **34** is coupled to a common central hub **36** which rotates about the rotation axis **18**. The hammers **34** are provided with bifurcated arms **38** which are pivotally coupled about respective bolts or pins **40** to the hub **36**. This enables the hammers **34** to swing if impacted by a hard foreign object which enters the impact zone **14** to minimise the likelihood of major damage. A hard foreign object, if not fragmented into pieces small enough to pass through the apertures **22**, will eventually exit through the gaps **24**.

[0132] Each hammer **34** has an outer axial edge **40** which extends for a length marginally smaller than the depth of the impact zone **14**. This enables the provision of a small clearance between the upper and lower radial edges of the hammers **34** and the annular plate **32** and bottom surface of the impact zone **14**.

[0133] The axial edge **40** is formed with a plurality of spaced apart grooves of flutes **44** the purpose of which is to assist in fragmenting elongated material such as straw that may enter the feed opening **12** as well as reduce smearing of material on the screen arrangement **20***a*. An impact side **46** of the hammers **34** is substantially planar and lies in the axial plane. A trailing face **48** of the hammers is scalloped. The purpose of this is to balance the impact mechanism **16** any radial plane. In this regard the hammers **34** extend in an axial direction higher than the hub **36**. In the absence of the scalloping the centre of gravity of the impact hammers **34** would be axially offset from the centre of gravity of the hub **36** which may lead to instability together with increased bearing wear and heat generation.

[0134] The combination of the impact mechanism **16** and the screen arrangement **20***a* forms a first milling stage of the multistage hammer mill **10**.

[0135] As can be seen from FIGS. **2** and **5***a* embodiments of the hammer mill **10** are provided with a first plurality of impact elements **50***a* disposed between the screen arrangements **20***a* and **20***b*. The combination of the first plurality of impact elements **50***a* and the second screen arrangement **20***b* forms a second milling stage of the multistage hammer **10**.

[0136] A second plurality of impact elements **50***b* is disposed between the screen arrangements **20***b* and **20***c*. The combination of the second plurality of impact elements **50***b* and the third screen arrangement **20***c* forms a third milling stage of the multistage hammer **10**.

[0137] The impact elements **50***a*, **50***b* (hereinafter referred to in general as “impact elements **50**”) between mutually adjacent screen arrangements are evenly spaced apart in the circumferential direction thus forming corresponding circular arrays of impacts elements. A lower end of each of the impact elements **50** is fixed a base plate **42**. An upper end of each of the impact elements **50***a* is attached to a ring **52***a*, while the upper end of each of the impact elements **50***b* is attached to a concentric ring **52***b*. The base plate **42** also forms the bottom surface of the impact zone **14**.

[0138] As shown on FIGS. **2** and **5***b*, each impact element **50** has a first flat surface **54** that lies parallel to the radial direction of the mill **10**. However in other embodiments the first flat face **54** may lie within 20 degrees to a radial direction of the multistage hammermill. Each impact element **50** also has on its radial inner side a second flat face **56** that joins, and forms an acute included angle with, the flat surface **54**. A curved (i.e. non-linear) surface **58** extends between the flat faces **54** and **56**.

[0139] The hub **36** and thus the central impact mechanism **16** are fixed to the base plate **42**. Thus the impact mechanism **16** and the impact elements **50** a driven together. When the impact elements are rotating about the rotation axis **18** the first flat face **54** is a leading face of the impact element **50** and provides for improved impact speeds. The curved surface **58** is a trailing surface and assists in reducing drag and turbulence. The second flat face **56** being at the acute angle relative to the first flat face **54** minimises sidewall impact of material moving radially outward's. This assists in improving airflow and chaff flow capacity.

[0140] The entire assembly of the base plate **42**, impact elements **50** and impact mechanism **16** may form a replaceable unit. Additionally, the flails **34** can be individually replaced by decoupling from the central hub **36**. Also, individual impact elements **50** or separate complete arrays of arrays of impact elements **50** may be replaceable.

[0141] The combination of the impact mechanism **16** and the impact elements **50** which are both attached to the base plate **42** forms a rotor structure **60**. The screen structure **33** inter-fits with the rotor structure **60** in a manner so that the annular plate **32** overlies the rings **52***a*, **50***b* and the base plate **42**; the first screen arrangement **20***a* locates between the hammers **34** and impact elements

50*a*; the second screen arrangement **20***b* interposes between the impact elements **50***a* and **50***b*; and the third screen arrangement **20***c* surrounds the impact elements **50***b*. A housing (shown in FIG. **13***b*) extends about the outer most screen arrangement **20** and is used to convert the pressure generated by the rotor into velocity at the exit. A discharge opening is formed in the housing. Material exits the multistage hammer mill through the discharge opening and is spread by the air flow generated initially by rotor structure **60** in particular the impact mechanism **16**.

[0142] If desired the screen structure **33** can also be driven to rotate about the rotation axis **18**. The screen structure **33** can be rotated in the same direction or in an opposite direction to the impact mechanism **16**/rotor structure **60**.

[0143] The general operation of the multistage hammer mill **10** is as follows. Material enters through the feed opening **12** and flows in the radial direction by airflow generated by the impact mechanism **16**. While in the primary impact zone **14** the material is accelerated by the hammers **34** and undergoes sheer, crushing, impact and attrition forces between the screen arrangement **20***a* and the hammers **34** multiple times. If the material is small enough to pass through the apertures **22***a* it passes to the next (second) milling stage constituted by the impact elements **50***a* and the second screen arrangement **20***b*. However, if the material isn't small enough, it has a maximum of approximately ⅓ rotation of the mill to reach an opening **24***a* where it subsequently passes to the second milling stage. In this way, over processing of material is prevented in an application where capacity is very important. As previously described above the number and/or relative position of the openings **24** can be adjusted to vary the maximum rotation.

[0144] Material in the second milling stage is impacted and accelerated by the impact elements **50***a* and pulverised against the screen arrangement **20***b*. Material that is small enough to pass through the apertures **22***b* enters the next (third) milling stage constituted by the impact elements **50***b* and the third screen arrangement **20***c*. Material that is not small enough passes into the third stage through an opening **24***b*.

[0145] Material in the third stage is impacted and accelerated by the impact elements **50***b* and pulverised against the screen arrangement **20***c*. Material that is small enough to pass through the apertures **22***c* enters a discharge chamber formed between the housing and the third screen arrangement **20***c*. Airflow in the discharge chamber exits together with entrained milled material through the discharge opening.

[0146] Embodiments of the disclosed multistage hammer mill have an advantage over traditional hammermills because reducing the screen size with each row allows smaller particles passing through quickly to the next stage. This reduces the amount of over pulverising on each row to improve the overall capacity of the mill for a given size.

[0147] Embodiments of the disclosed hammermill approach are believed to have an advantage over the Berry Saunders mill by virtue of the screen arrangements **20** enabling control over particle size. In particular screen arrangements **20** of different aperture **22** sizes can be interchanged to facilitate adjustment to target different weed species. Additionally, the screen arrangements **20** are radially narrow and therefore rotating impact elements **50** can be close together radially and operate at similar tip speeds. It is believed that the impact elements operating at similar tip speeds improve seed kill effectiveness and energy efficiency. Additionally, the multistage hammer mill is able to provide shear, crushing and attrition to material for more effective processing of fibrous crop materials.

[0148] In one embodiment the output airflow and chaff material can be used to assist the spread of a straw chopper by directing onto the chopper tailboard, which has either stationary vanes or rotating spinners or otherwise to spread the residue material.

[0149] In another embodiment the output of the material from the disclosed mill can be directed into a straw chopper itself. By combining chopper and the multistage hammermill air flows the overall performance can be improved. For example, the chopper and multistage hammermills will require a certain amount of air flow operating individually to process and distribute residue material. By operating in series, this amount of air flow pumping could be reduced and still be able to process and distribute material effectively. This could be achieved by reducing the air flow effect of either or both of the chopper and impact mill.

[0150] FIGS. **7***a* and **7***b* illustrates a part of a residue processing system **80** which comprises at least one but in this case two multistage hammer mills **10***a* and **10***b* in a side-by-side juxtaposition. The residue destruction system **80** may also include a chopper (not shown) arranged relative to the hammer mills as described above. The hammer mills **10***a* and **10***b* (hereinafter referred to in general as "hammer mills **10**") in this embodiment are of the same structure and design as the hammer mill **10**.

[0151] The residue processing system **80** includes a drive system **82** for driving the hammer mills **10**. The drive system **82** has a main pulley **84** for driving a first belt **86** and a second belt **88**. The first belt **86** runs about an idler **90**, a drive pulley **92***a* and a drive pulley **92***b*. An outer surface of the belt **86** drives the pulley **92***a* while an inner side of the belt **86** drives the pulley **92***b*. As a consequence, the pulleys **92***a* and **92***b* rotate in mutually opposite directions. The pulley **92***a* imparts torque to a drive shaft **93***a* of the impact mechanism **16** and the corresponding rotor structure **60** of the mill **10***a*. The pulley **92***b* imparts torque to a drive shaft **93***b* the impact mechanism **16** and the corresponding rotor structure **60** of the mill **10***b*.

[0152] The second belt **88** runs about an idler **94** and drive pulleys **96***a* and **96***b*. An outer surface of the belt **88** drives the pulley **96***b* while an inner side of belt **88** drives the pulley **96***a*. Accordingly the pulleys **96***a* and **96***b* rotate in mutually opposite directions. The pulley **96***a* imparts torque to a drive shaft **95***a* of the screen structure **33** of the mill **10***a* while the pulley **92***b* imparts torque to a drive shaft **95***b* the screen structure **33** of the mill **10***b*.

[0153] It should be recognised that the pulleys **92***a* and **96***a* rotate in mutually opposite directions; as do the pulleys **92***b* and **96***b*. Thus, the drive system **82** operates to drive the rotor structures **60** and screen structures **33** for each hammer mill **10** in mutually opposite directions.

[0154] The main pulley **84** is coupled to a transmission system **98**. In the present illustrated embodiment, the transmission system **98** comprises a pulley **100** which is coupled by shaft **102** to a gearbox **104** which has an output shaft **106** that drives the pulley **84**. The pulley **100** is driven by a belt **108** which receives power from a power source (not shown) that drives the belt **108** about a power axis that is perpendicular to the shaft **106** and to the rotation axes of the shafts **93***a*, **93***b*, **95***a* and **95***b*. The use of drive belts **86** and **88** to impart torque to the hammer mills **10** assists in reducing shock loads on the gearbox **104**.

[0155] The residue processing system **80** may be part of an agricultural machine such as but not limited to a combine harvester. FIGS. **8***a* and **8***b* are schematic representations of a rear portion of a combine harvester **120** depicting a chopper **122** with radial chopper blades **123**, a tailboard **124** and fitted with two multistage hammer mills **10***a* and **10***b* (hereinafter referred to in general as "hammer mills **10**"). The chopper **122** is driven to rotate about an axis **126** which is parallel to a power take off shaft (not shown) of the combine harvester **120**. The power take off shaft extends in a direction perpendicular to the direction of travel of the combine harvester **120**.

[0156] The hammer mills **10** are driven by the drive system **82** which is also powered by the power take off shaft of the combine harvester **120**. Specifically, the belt **108** is engaged with a pulley (not shown) mounted on the power take off shaft. It should be appreciated here that the hammer mills **10** are mounted in a manner so that their respective impact mechanisms **16** are rotated about axes that are perpendicular to the power take off shaft and the axis **126**. In the arrangement shown in FIGS. **8***a* and **8***b* mills **10** are arranged so that their discharge flow is directed or otherwise fed into the chopper **122**. Thus, the airflow of the hammer mills **10** is added to the airflow of the chopper **122** which may provide a synergistic effect. More particularly the airflow from the hammer mills **10** may be added to the respective axial end regions of the chopper **122**. This may assist in providing greater sideways or lateral spreading of the material from the chopper **122**. This effect may be further enhanced by installing curved blades or fins **125** in an outlet chute **127** of the chopper **122** at least near or adjacent its axial end regions.

[0157] In the arrangement shown in FIG. **9** the discharge flow from the hammer mills **10** is directed onto the tailboard **124** of the chopper **122** to assist in spreading their respective discharged processed materials.

[0158] FIGS. **10***a* and **10***b* show an alternative form of drive system **82***a* for transferring drive from a power take off shaft **130** of the combine harvester **120** to the hammer mills **10***a* and **10***b*. In these Figures the same reference numbers are used to denote the same features as described for the system **82** shown in FIGS. **7***a* and **7***b*.

[0159] The drive system **82***a* has many similarities to the drive system **82** shown in FIGS. **7***a* and **7***b* in that it includes the gearbox **104** driven by the PTO **130** via the belt **108** and pulley **100**; and the gear box **104** rotates a pulley **84** that drives the hammer mills **10***a* and **10***b*. However, the drive system **82***a* also includes a PTO shaft **132** connected between the gear box driveshaft **106** and the drive pulley **84**. The drive pulley **84** drives to belts **86** and **88**. The belt **86** engages the pulley **92***a* to drive the driveshaft **93***a* for the impact mechanism of the mill **10***a*. The drive belt **88** engages the pulley **92***b* to drive the driveshaft **93***b* for the impact mechanism **16** of the mill **10***b*. An idler pulley **90** is provided to enable tension variation in the belt **88**. By this arrangement the shafts **93***a* and **93***b* are driven in the same direction but the screen arrangements **20** of the mills **10** are not driven, rather they remain stationary.

[0160] FIGS. **11***a* and **11***b* show yet a further variation of the drive system **82***b*, in which the same reference numbers are used to denote the same features of the drive system **82** shown in FIGS. **10***a* and **10***b*. In the system **82***b* the impact mechanisms **16** of the hammer mills **10** are driven by a single belt **89** which engages the pulley **84**, the idler pulley **90** and pulleys **92***a* and **92***b*. The gearbox **104** receives power from the harvester PTO **130** by a two belts **108***a* and **108***b* and an intervening jack shaft **134**.

[0161] FIG. **12** shows a further drive system **82***c* which is somewhat of a hybrid between the system shown in FIGS. **10***a* and **11***a*. In the system **82***c* drivers received from a belt **108***b* (from FIG. **11***a*) to drive pulley **100** coupled to a gearbox (not visible in FIG. **12**). The gearbox drives the pulley **84** to rotate about an axis perpendicular to that of the pulley **100**. The pulley **84** drives belts **86** and **88**. These belts engage with pulleys **92***a* and **92***b* of corresponding hammer mills **10**. Due to the drive arrangement the hammer mills **10** are driven in the same direction as each other. Idler pullies **91** are provided for tensioning the belts **86** and **88**.

[0162] In the drive system **82***c* a fan **140** is optionally incorporated in the pulley **84**. The pulley **84** is formed with a belt engaging ring **142**, a central hub **144** and a plurality of pitched fan blades **146** emanating from the hub **144** to the inside of the ring **142**. In this way the pulley **84** acts as a cooling system for the gearbox to which it is connected. It should be appreciated that other pulleys described in earlier drive systems may also incorporate a similar fan to provide cooling to gearboxes or indeed other parts and components including the hammer mills **10** themselves. For example, the pulleys **100** shown in FIGS. **7***a* and **10***a* can be formed with fans **100**.

[0163] It will be also recognised that in each of the described residue processing systems **80**, drive/torque from the PTO **130** is transmitted through 90° to rotate shafts **92**, **93**. The shafts **92** and **93** are shown extend in a vertical plane when mounted on a harvester **120**, but could be slanted towards the front of the harvester or towards the rear of the harvester.

[0164] FIGS. **13***a* and **13***b* depict a possible orientation and juxtaposition of two hammer mills **10** when rotated in the same direction and mounted on a combine harvester. FIG. **13***a* shows the hammer mills **10** with their respective covers **148** on, while FIG. **13***b* shows the same arrangement but with the covers **148** off. Here both of the hammer mills **10**, and more particularly the impact mechanisms **16** are rotated in a clockwise direction as indicated by the arrows drawn on the respective upper annular plates **32**. The hammer mills **10** are mounted on a common base plate **150**. Each base plate has a substantially circular portion **152** and an outlet portion **154**. The hammer mills **10** are eccentrically mounted on the respective circular portions **152** so that a radial distance

156 between the outer peripheral radius of the hammer mills **10** and the edges of the circular portions **152** increases in the direction of rotation toward the outlet portion is **154**. This assists in airflow through the hammer mills **10**.

[0165] Whilst a number of specific embodiments of the mill and residue processing system have been described, it should be appreciated that the mill and system may be embodied in many other forms. For example, the illustrated embodiment shows a three stage hammer mill with respective screen arrangements **20** each having apertures **22** of progressively smaller dimension with distance away from the rotation axis **18**. However, in one embodiment the size of the apertures **22** can be the same for all of the screen arrangements **20**. Alternately the size the apertures **22** can be arranged so that the size stays the same or decreases with increased radius from the rotation axis **18** relative to the aperture size of a radially inward adjacent screen arrangement **20**. In yet a further variation the orientation of the apertures may vary between respective screen arrangements. For example, the apertures **22***a* may be of a rectangular shape having a major axis parallel to the rotation axis, while the apertures **22***b* may be of the same size and shape of apertures **22***a* but orientated so that their major axis is +45° to the rotation axis **18**, and apertures **22***c* again of the same size and shape but orientated so that their major axis is −45° to the rotation axis **18**.

[0166] In other variations the mill **10** may be formed with screen arrangements **20** that have either: no gaps **24**; or one or more gaps in the inner most screen arrangement **20***a* and either no or one or more gaps in radially outer screen arrangements. Also, while the illustrated embodiment shows gaps **24** in successive screen arrangements **20** having some degree of overlap, in other embodiments the gaps in respective screen arrangements may be offset from each other so as to not overlap.

[0167] In each of the illustrated embodiments of the hammer mill **10** the first screen arrangement **20***a* is radially adjacent the central impact mechanism **16** and associated flails/hammers **34**. However this is not an essential requirement. One or more circumferential arrays of impact elements (for example similar to the impact elements **50**) may be interposed between the impact mechanism **16** and the first screen arrangement **20***a*. This is exemplified in FIG. **14** which shows embodiment of the multistage hammer mill **10***c* having: an impact mechanism **16** with flails/hammers **34** rotatable about a rotation axis **18**; a first screen arrangement **20***a*; a second screen arrangement **20***b*; and a first plurality of impact elements **50***a* is disposed between the screen arrangements **20***a* and **20***b*; as per each of the earlier described embodiments of the hammer mill **10**. However the hammer mill **10***c* also includes two circumferential arrays A**1** and A**2** of impact elements **50**. The radially inner array A**1** of impact elements **50** may be: stationary; arranged to rotate in the same direction as the impact mechanism **16**; or, arranged to rotate in an opposite direction to the impact mechanism **16**. The radially outer array A**2** of elements **50** may be arranged to rotate with the impact mechanism **16**. In a modified form of the hammer mill **10***c*, the radial inner array A**1** of impact elements **50** may be dispensed with so that the modified hammer mill **10***c* includes only the rotating array A**2** interposed between the impact element **16** and the first screen arrangement **20***a*. In these embodiments the impact mechanism **18**, the arrays of impact elements A**1**, A**2** and the first screen arrangement **20***a* make up the first hammer mill stage.

[0168] FIGS. **15***a* and **15***b* show further possible modification to the screen segments **26** which make up the respective screen arrangements **20**. Here a plurality of ribs **28***f* is fixed to a radial inner side of the screen segments **26**. The ribs **28***f* extend in the axial direction and are circumferentially spaced apart. Conveniently respective ribs **28***f* are located in the space between mutually adjacent columns of apertures **22**. The addition of ribs **28***f* slow the material traveling around the screen arrangements **20**, keeping the material in the impact zone for longer and thereby increasing the shear and impact forces on the material. Any one of the screen arrangements **20** can be provided with one or more of the ribs **28***f*.

[0169] FIG. **16** illustrates a modified or alternate form of the screen arrangements **20***a*′, **20***b*′ and **20***c*′ (hereinafter referred to in general as screen arrangements **20**′). The substantive differences between the screen arrangements **20**′ and the screen arrangements **20** are as follows. In the screen arrangements **20**′ upper rings **30***au*, **30***bu* for the screen arrangements **20***a*′, **20***b*′ extend laterally from a radial outer side of the respective screen arrangements to a location close to (but not touching) a radial inner side of the screen arrangements **20***b*′ and **20***c*′ respectively. This avoids the creation of a substantial gap between the upper surface of the rings **52** and the inside surface of the annular plate **32**. By way of comparison phantom line F in this Figure shows the location of the radial outer side of the upper rings **30***b* and **30***c*.

[0170] Additionally, in screen arrangements **20**′ the apertures **22** include an uppermost row apertures **22***u*, for at least the second and third milling stages, that extend in the axial direction to at least an under surface of the upper rings **30***au* and **30***bu*. A benefit of this arrangement is that material located in a region R between the inside of the annular plate **32** and the rings **52** can pass through the apertures **22***u* to the next milling stage. This minimises the risk of material building up in the region R.

[0171] FIG. **17** illustrates further possible variations of the disclosed hammer mill **10** in which axial and radial scrapers **51** and **53** respectively, are associated with the impact members **50**. This association is by way of the scrappers being provided on the rings **52***a* and **52***b* of the corresponding circular array of impact members. The axial scrapers **51** are formed on the upper surfaces **54***a* and **54***b* of the corresponding rings. The scrapers **51** act to clear material in the regions R and assist in directing that material to pass through the apertures **22***u*. The scrapers **53** are formed on a radial outer circumferential edge of the rings **52***a* and **52***b* and extended to a location close to but not touching the adjacent screen arrangements **20**, **20**′. The purpose of the scrapers **53** is to also assist in directing material to pass through the apertures **22***u*. Moreover the scrapers **53** assist in preventing a build-up of material between the rings **52***a*, **52***b* and the adjacent screen arrangements.

[0172] FIG. **18** shows another modification where a screen segment **26** adjacent one of the openings **24** of a screen arrangement **20** is replaced with a pulverising block **160**. The pulverising block **160** has a solid front face formed with a sawtooth like profile. The block **160** provides an additional grinding and crushing zone within a milling stage. More than one block **160** can be incorporated in each milling stage. For example one screen segment **26** immediately adjacent an opening **24** could be replaced with a block **160**. For a screen arrangement having three openings **24** there would then be three blocks **160**. Ideally each block **160** would be on a leading side of the opening **24** with reference to the direction of rotation of the corresponding impact elements **50**.

[0173] FIG. **19** shows is a further slight modification or variation in which the ribs **28** that would otherwise be on adjacent sides of an openings **24** are replaced with plates **28***p* that are angled in the direction of rotation of the impact mechanism **16** and impact elements **50**. This would serve to increase the velocity of material and air exiting the screen arrangement for increased capacity. This may be particularly beneficial for the outermost milling zone/screen arrangement **20**.

[0174] In a further variation the cross sectional shape of the impact elements **50** may be varied for that specifically shown in FIGS. **2** and **5***a* and **5***b*. For example, the impact elements **50** may have a simple circular cross-sectional shape.

[0175] Embodiments of the disclosed multistage hammer mill **10** have a minimum of two milling stages. The embodiment described and illustrated in the present drawings is provided with an optional third milling stage. It should be understood however that additional milling stages can be sequentially added with increased radius from the rotation axis **18**, each additional milling stage comprising a screen arrangement and an array of impact elements **50**. It is also possible in one embodiment for the milling stages to be arranged so that material milled in the first milling stage passes through at least one subsequent adjacent milling stage, or alternately through all of subsequent milling stages.

[0176] As previously described the provision of the openings **24** in the screen arrangements **20** is an optional feature. In one variation an embodiment of the hammer mill **10** may be formed in which the first milling stage is formed with no openings **24** in the first screen arrangement **20***a*. In this way hard materials are prevented from passing through sequential milling stages and into possible other mechanisms in a harvester such as a chopper. In such a variation the hammer mill **10** may also be provided with one or more sensors and an alarm to notify an operator of the existence of hard materials circulating within the first milling stage.

[0177] Weed seeds and crop residue material have varying properties. The amount of destruction (i.e. crushing, shearing, impact and attrition) needed depends on the seeds being targeted and the residue material that is being processed. Embodiments of the disclosed multistage hammer mill **10** enable the degree of destruction of residue material to be increased by: [0178] 1) increasing the relative rotational speed to increase impact and shear forces; [0179] 2) reducing the size of the screen openings **22** to keep larger material in the impact zone for longer; [0180] 3) increasing the circumferential spacing of the openings **24** allowing larger material to be processed for longer before passing through; [0181] 4) providing the inner ribs **28***f* to increase residence time in the impact zones.

[0182] In a variation to the above described drive system **82** the main drive **98** may be in the form of a hydraulic pump powered by the PTO **130** which provides hydraulic fluid to a hydraulic motor coupled which drives the shaft **106**. This avoids the need for the gearbox **100**. A potential benefit in using the hydraulic motor is better speed control and the inherent ability to provide a soft start. This method is believed to be more efficient than directly driving two mills individually as it requires only one hydraulic motor which can be operated at optimum speed (slower) and pressure.

[0183] It should also be understood that when the residue processing system **80** or the combine harvester **120** has only a single residue processing device the corresponding drive system **82** is simplified by requiring: only a single drive belt drive and a single shaft in the event that the residue processing system has only one rotary component. In the event that the single residue processing device has counter rotating components then two belts will be required however the number of pulleys required to be driven is reduced in comparison to the above described processing systems and combine harvesters having two or more side-by-side residue processing devices.

[0184] Also in the above-described residue processing systems **80** the residue processing devices are exemplified by embodiments of the disclosed hammer mill **10**. However, the residue processing system **80** may use different types of residue processing devices such as but not limited to, pin mills, cage mills single stage hammermills, chaff spreaders and straw choppers. That is, the residue processing system **80** and the associated drive system **82** is independent of the specific type of residue processing device.

[0185] As previously described embodiments of the disclosed hammer mill **10** and the residue processing system **80** may incorporate various sensors for the purposes of monitoring performance and providing status information. The output from such sensors may be used to cause the generation of signals or other indication of impending faults and thereby alert an operator of the need for maintenance or indeed immediate shutdown to avoid substantial and/or catastrophic failure of the hammer mill **10** or processing system **80**.

[0186] For example, the vibration sensors V**1**, V**2** shown in FIG. **7***a* may be incorporated in the hammer mill **10** to give provide signals pertaining a measure of vibration about the axis of the shafts **95***a* and **95***b*. The mill **10** may incorporate or can otherwise be associated with. a data processor (not shown) for example on an associated combine harvester or other machine/vehicle on which the mill **10** and processing system **80** are amounted. The data processor may be arranged to process the vibration sensor signals and provide feedback to the user of the condition of the mill **10** or any other high speed rotating device. The data processor may be arranged to process signals from the vibration sensors and: (a) issue a warning or alarm to an operator of the mill; or, (b) facilitate an autonomous reduction in rotational speed of the mill; or, (c) facilitate a shut down the mill to a rotational speed of 0 rpm; or (a) and (b); or (a) and (c); if the processed signal are indicative of mill vibration being in excess of a threshold. For example, the data processor may execute one or more algorithms to determine an out of specification vibration reading may provide

early detection warning of worn, damaged or out of balance components.
[0187] In addition, or alternately, other sensors may be incorporated including but not limited to:
[0188] Blockage sensors which may detect blockage in or reduced mass flow rate of material through the mill. The blockage sensors can for example be placed to detect (a) a blockage at or reduced mass flow into the central feed opening **12** of the mill **10**; and/or (b) a blockage at or reduced mass flow rate from the outlets **154** of the mill **10**. Examples of such sensors B**1**, B**2**, B**3** and B**4** (in general “blockage sensors Bj” where j=1, 2, 3, 4, etc) are shown in FIG. **13***a*. [0189] Temperature sensors to ensure the safe operation of gearboxes and/or bearings. [0190] Temperature measurements from the temperature sensors may be fed to the data processor. The data processor may be arranged to issue an alarm or warning to an operator of the mill of the temperature measurement being above a threshold. The data processor may alternately or additionally autonomously reduce the speed or indeed stop the will rotation of the mill when the sensed temperature is over a threshold limit. Additionally, or alternately the mill may be provided with an optional cooling system which is autonomously activated by the data processor to cool the bearing when the temperature measurements exceed the threshold. Activation of the cooling system may be done conjunction with one or both of issuing a warning/alarm and reducing the speed or stopping rotation of the impact mechanism. [0191] One or more proximity sensors may be incorporated in or otherwise associated with the impact mill **10** to monitor the rotation speed of the mill **10** and provide or cause the generation of an alarm when a person is within a designated distance of the mill **10**. Such sensors may also be arranged to directly or via the data processor to activate a light and/or an audible signal generator to indicated or otherwise show that the mill **10** is still rotating thereby alert the person that machine is not safe to work on. [0192] One or more torque sensors Ti sensors (see FIG. **10***a*) which are arranged to provide a signal indicative of torque or load applied to the drive shaft **93***a*, **93***b* of the impact mechanism. In one embodiment each torque sensor may comprise first and second encoders E**1**, E**2** spaced apart along with the drive shaft **93***a*, **93***b*. In such embodiment the data processor may be programmed or otherwise arranged to calculate or provide a measure of torque on the basis of a difference between outputs of the first and second encoders. In one embodiment the first and second encoders are located at opposite ends of the drive shaft.
[0193] In each of the above described embodiments the data processor associated with the mill and/or the combine harvester may also include or be coupled with a communications system enabling data, signals or information from any one or more of the sensors (i.e. the vibration sensors, blockage sensors, temperature sensors, proximity sensors, and torque sensors) to be communicated via a communications network including but not limited to the Internet or the Internet of things, to a remote location. The data, signals or information from the sensors may be provided directly from the sensors, or, as processed data, signals or information subsequent to processing by the data processor, or both.
[0194] Communicating the data, signals or information enables remote monitoring of the performance of the mill and/or the combine harvester. The remote monitoring can for example enable manual or automated communication to a combine operator or a service department of performance characteristics of the mill and/or the combine harvester. The performance characteristics may include: information regarding wear of various components, the need for maintenance, or the provision in real time of alerts or alarms to the combine operator of potentially dangerous performance characteristics such as bearing temperature.
[0195] The data, signals or information may also be used, together with other operational information communicated via the communication system such as forward speed of the combine harvester, to calculate the amount of material processed by the combine harvester. This may be beneficial in terms of different business or revenue models for commercialisation of the mill and/or combine harvester in enabling for example lease payments/charges being made on the basis of the calculated amount of material processed by the combine harvester.
[0196] Therefore, embodiments of the disclosed mill may include any one or any combination of

two or more of the vibration sensors V, blockage sensors Bj, temperature sensors, torque sensors Tn and proximity sensors. Data, signals or information from the sensors can be processed on-board the combine harvester using the data processor, or can be transmitted for processing and analysis at a remote location. The data, signals or information can be used in real time or otherwise to: [0197] to alert the combine harvester of various operational characteristics of the mill and/or the harvester; [0198] signal the need for maintenance and/or repair; [0199] determine performance characteristics of the mill and/or the combine harvester; [0200] enable calculation of throughput of material processed by the mill and/or the combine harvester.

[0201] The data, signals and information communicated to the remote location may be stored locally or on a cloud-based system. In any event the data, signals and information may be fed to a machine learning/artificial intelligence system. This in turn may be arranged for example to: forecast expected lifespan of components, machine throughput; and/or suggest potential adjustments to mill or combine harvester parameters to improve operational efficiency.

[0202] The blockage sensors Bj exemplified above may be also be incorporated in the combine harvester to assist in monitoring the natural flow of material through or out of the harvester. Some harvesters have mechanisms to mechanically move the crop and residue material, others rely on steep sides and gravity. Crop materials vary massively between different crop types, moisture conditions and presence of weeds. Often these materials can be green, sappy and difficult to flow. Blockages in residue processing devices are a significant issue for many operators of combines harvesters. There is very little feedback to determine if a blockage has occurred until the harvester is overloaded with crop residues. This can often result in machine damage or at a minimum down time while the operator needs to remove the residues. Having early detection of blockages allows the operator to stop quickly to prevent the whole machine clogging and may even allow the operator to slow down and allow the blockage to clear.

[0203] In one non-limiting example the blockage sensors Bj may be in the form of a capacitive sensor. The sensors Bj can detect a build-up of crop residue for the prevention of blockages. Preferably one or more of the sensor Bj are attached to a crop residue processing device such as the mill **10** with an associated instrument mounted in the driver's cab to provide the operator with visual and/or audible indication of blockage status. Alternately or additionally one or more blockage sensors Bj can be mounted on the feed chutes of a crop residue processing device.

[0204] The blockage sensors Bj may be operatively coupled to the data processor which can be programmed to take one or more specific actions if a blockage is detected or an anomalous change in material flow is detected. These actions may include but are not limited to: operating a high-pressure air compressor to direct one of jets of air to a location where the change in material flow or blockage is detected; and/or reducing the ground speed of a machine (e.g. a combine harvester) to reduce the volume of material being directed to the blockage site. In the latter case the data processor may alert an operator if and when that the blockage has cleared to enable a resumption of normal travel speed.

[0205] When the mill is on a combine harvester **120** with a chopper **122** that if fed with the residue from the discharge of the mill, one or more additional blockage sensors Bj can be arranged detect a blockage in or reduced mass flow rate of material through the chopper **122**.

[0206] The above described vibration, blockage, temperature and proximity sensors; and the data processor and its above described functionality with these sensors are not limited to application with the disclosed multistage hammer mill and combine harvester. The sensors and data processor may be used with or otherwise incorporated in other impact mills and rotary residue processing equipment including for the example those described in the Background Art on pages 1-4 of this specification. Impact mills use high impact speeds generated by one or more rotating elements (such as an impact mechanism **16** and/or rotating elements **50**) which rotate about a rotation axis to pulverise the material. The rotating elements rotate on a bearing and the impact mill requires an inlet for material to enter the mill and an outlet for discharge of the milled material

[0207] In the claims which follow, and in the preceding description, except where the context requires otherwise due to express language or necessary implication, the word “comprise” and variations such as “comprises” or “comprising” are used in an inclusive sense, i.e. to specify the presence of the stated features but not to preclude the presence or addition of further features in various embodiments of the hammer mill and residue processing system as disclosed herein.

Claims

1. A combine harvester comprising at least one impact mill for devitalising weed seeds in a crop while the crop is being harvested by the combine harvester wherein the combine harvester has a power take off shaft and the impact mill includes: an inlet for material to enter the mill, an impact mechanism arranged to rotate about a rotation axis, the impact mechanism being operable to impact the material after entering through the inlet and accelerate the impacted material in a radial outward direction, and an outlet for discharge of impacted material in a radial direction relative to the rotation axis; drive system coupled to the power take off shaft, the drive system arranged to transfer torque from the power take off to the impact mechanism; one or more torque sensors arranged to provide a signal indicative of torque applied to a drive shaft of the impact mechanism, and a data processor arranged to process information provided by the one or more torque sensors wherein the data processor is programmed to calculate throughput of material processed by the impact mill.
2. The combine harvester according to claim 1 including one or more vibration sensors arranged to sense vibrations arising from rotation of the impact mechanism about the rotation axis, and wherein the data processor is arranged to process information provided by the one or more vibration sensors and the one or more torque sensors to calculate throughput of material processed by the impact mill.
3. The combine harvester according to claim 1 including one or more blockage sensors arranged to detect blockage in, or reduced mass flow rate of material through, the mill, and wherein data processor is arranged to process the information provided by two or more of (a) at least one torque sensors, and (b) at least one vibration sensor, and (c) at least one blockage sensor to calculate throughput of material processed by the at least one impact mill.
4. The combine harvester according to claim 1 including at least one temperature sensor arranged to provide a temperature measurement of a bearing on which the impact mechanism rotates wherein the data processor is arranged to process the information provided by the one or more temperature sensors in addition to the at least one of the torque sensors to calculate throughput of material processed by the at least one impact mill.
5. The combine harvester according to claim 3 including one or more high-pressure air jets capable of directing jets of high pressure air to one or both of the mill inlet and the mill outlet and wherein the data processor is arranged to cause the one or more high-pressure air jets to direct jets of high pressure air to one or both of the mill inlet when an anomalous change in material flow or a blockage is detected.
6. The combine harvester according to claim 1 wherein the data processor is arranged to determine performance characteristics of the mill and/or the combine harvester on the basis of information received from one or more of the one sensors.
7. The combine harvester according to any one of claim 1 further comprising: a chopper arranged to receive a discharge flow from the impact mill; and one or more blockage sensors arranged detect a blockage in or reduced mass flow rate of material through the chopper.
8. The combine harvester according to claim 4 wherein the data processor is arranged to: (1) issue an alarm or warning to an operator of the impact mill of the temperature measurement being above a threshold; or (2) autonomously reduce the speed or stop the rotation of the impact mechanism; or (3) activate a mill cooling system for cooling the bearing; or (4) any combination of two or more of (1), (2) and (3); when the temperature measurements exceed the threshold.

9. A combine harvester comprising: at least one impact mill and a straw chopper, wherein the impact mill is capable of one or both of (a) discharging material and air into the straw chopper or (b) discharging material and air in a manner to add to airflow of the chopper; the at least one impact mill comprising an inlet for material to enter the mill, an impact mechanism arranged to rotate about a rotation axis, the impact mechanism being operable to impact the material after entering through the inlet and accelerate the impacted material in a radial outward direction, and an outlet for discharge of pulverised material in a direction of rotation; and (a) one or more torque sensors arranged to provide a signal indicative of torque applied to a drive shaft of the impact mechanism; or (b) one or more blockage sensors arranged detect blockage in or reduced mass flow rate of material through the mill and one or more torque sensors arranged to provide a signal indicative of torque applied to a drive shaft of the impact mechanism.
10. The combine harvester according to claim 9 comprising one or more additional blockage sensors arranged detect a blockage in or reduced mass flow rate of material through the chopper.
11. The combine harvester according to claim 9 wherein the at least one impact mill comprises two impact mills each being capable of one or both of (a) discharging material and air into the straw chopper, or (b) discharging material and air in a manner to add to airflow of the chopper.
12. An impact mill system capable of mounting on a combine harvester for devitalising weed seeds in a harvested crop while the crop is being harvested by the combine harvester, the combine harvester having a power take off shaft, the impact mill system comprising: an impact mill having an inlet for material to enter the mill, an impact mechanism arranged to rotate about a rotation axis, the impact mechanism being operable to impact the material after entering through the inlet and accelerate the impacted material in a radial outward direction, and an outlet for discharge of the impacted material in a radial direction relative to the rotation axis; a drive system coupled to the power take off shaft, the drive system arranged to transfer torque from the power take off to the impact mechanism; one or more torque sensors arranged to provide a signal indicative of torque applied to a drive shaft of the impact mechanism, and a data processor arranged to process information provided by the one or more torque sensors wherein the data processor is programmed to calculate throughput of material processed by the impact mill.